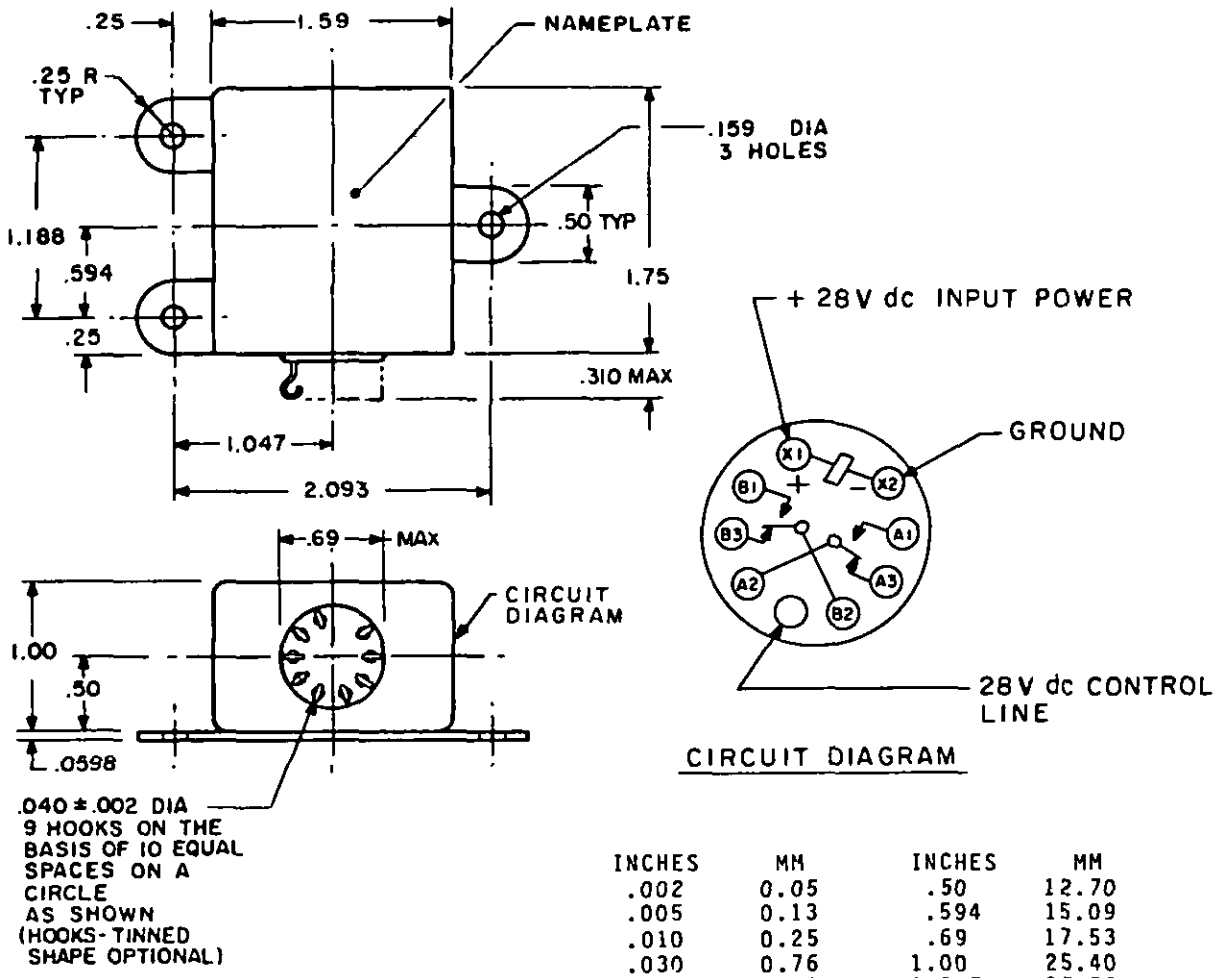
MILITARY SPECIFICATION SHEET

Ⓑ RELAYS, HYBRID, TIME DELAY (ON RELEASE), TYPE IIA, CLASS A, 2 AMPERES, 2PDT, FIXED TIME, 0.100 TO 300 SECONDS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83726.

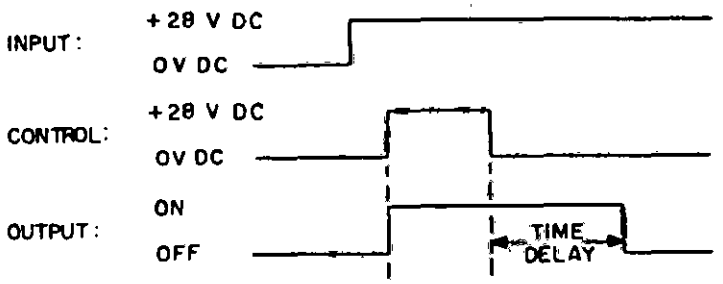


INCHES	MM	INCHES	MM
.002	0.05	.50	12.70
.005	0.13	.594	15.09
.010	0.25	.69	17.53
.030	0.76	1.00	25.40
.040	1.02	1.047	26.59
.0598	1.52	1.188	30.18
.159	4.04	1.59	40.39
.25	6.35	1.73	44.45
.310	7.87	2.093	53.16

Ⓑ FIGURE 1. Outline dimensions and configuration of relay.

Ⓑ denotes changes

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TIMING DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Metric equivalents are given for general information only.
(B) 4. Terminal numbers shall not appear on the relay header and there shall be affixed to the relay a permanent legible schematic diagram that identifies each terminal location specified.

FIGURE 1. Outline dimensions and configuration of relay - Continued.

(B) TABLE I. Available time delay relays. 1/ 3/

Dash number	Time delay seconds ±10%	Dash number	Time delay seconds ±10%	Dash number	Time delay seconds ±10%	Dash number	Time delay seconds ±10%
-1000	2/ .100	-1402	14	-7002	70	-1803	180
-5000	2/ .500	-1602	16	-7502	75	-1903	190
-7500	2/ .750	-1802	18	-8002	80	-2003	200
-1001	1	-2002	20	-8502	85	-2103	210
-2001	2	-2202	22	-9002	90	-2203	220
-3001	3	-2502	25	-9502	95	-2303	230
-4001	4	-3002	30	-1003	100	-2403	240
-5001	5	-3502	35	-1103	110	-2503	250
-6001	6	-4002	40	-1203	120	-2603	260
-7001	7	-4502	45	-1303	130	-2703	270
-8001	8	-5002	50	-1403	140	-2803	280
-9001	9	-5502	55	-1503	150	-2903	290
-1002	10	-6002	60	-1603	160	-3003	300
-1202	12	-6502	65	-1703	170		

(B) 1/ Additional time delay relays within the .100- to 300-second delay range are available. To establish part numbers not listed in table I, refer to page 5.
(B) 2/ Add ±10 milliseconds to ±10 percent tolerance.
(B) 3/ The suffix letter W, X, or Y to designate quality level shall be added to the dash number, refer to page 5.

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REQUIREMENTS:

CONTACT DATA:

Configuration: 2PDT; 2 form C.

Life/load ratings (relay case grounded):

Type of load	Life (cycles)	Amperes 28 V dc	Amperes 115 V ac - 1 phase 60 and 400 Hz
Resistive	100,000	2.0	0.3
Inductive	100,000	1.0	0.3
Lamp	100,000	0.1	0.1

Contact voltage drop:

Initial: 0.150 volt.

After life tests: 0.200 volt.

Minimum current: Applicable per MIL-R-5757.

Contact bounce: 2 milliseconds maximum, to be measured on each contact set using an oscilloscope, contacts shall be loaded with rated current and voltage.

Overload:

Resistive: 200% rated current.

Inductive: 200% rated current.

INPUT DATA:

Duty rating: Continuous.

Range of voltage:

Input power: 18 to 31 V dc.

Control line power: 18 to 31 V dc.

Minimum voltage - high temperature: 18 V dc.

Maximum current at 25°C and 28 V dc: 110 milliamperes.

Time delay: Fixed time .1 to 300 seconds (see table I).

Time delay tolerance: ±10% except as noted in table I, under all combinations of input voltage and all environmental conditions.

Recycle characteristics:

Before time out: Control power must remain off at least 25 milliseconds or 1% of the nominal time delay, whichever is greater, after which reapplication of control power for 25 milliseconds minimum will recycle the timer with a loss in timing no greater than 10%.

After time out: Reapplication and subsequent removal of control power will recycle the timer. Control power must be applied for 25 milliseconds or 1% of the total time delay, whichever is greater, to insure a loss in timing accuracy no greater than 10%.

Polarity protection: The unit will be inoperative during and undamaged by, reversal of the input polarity.

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ELECTRICAL DATA:

Insulation resistance at 500 V dc 1/:

Initial: 1,000 megohms minimum.

After life or environmental tests: 500 megohms minimum.

Dielectric strength (sea level):

Input (1 and 9) 1/:

Initial: 1,000 V rms.

After life: 1,000 V rms.

All other points 2/:

Initial: 1,000 V rms.

After life: 500 V rms.

Dielectric strength (altitude):

Input (1 and 9): 350 V rms. 1/

All other points: 350 V rms. 2/

Transients: MIL-STD-704A, figure 9, limit 1 for category B equipment.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Altitude: 80,000 feet.

Shock: 50 G's for 11 ±1 milliseconds, MIL-STD-202, method 213A, condition A.

Contact opening: 10 microseconds maximum duration. Monitor per method 310 of MIL-STD-202.

Vibration (sinusoidal): 10-80 Hz at 0.06 inch peak double amplitude, 80-3,000 Hz at 20 G's.

Acceleration: 50 G's steady state, no opening of closed contacts.

(8) Sealing: Not applicable.

PHYSICAL DATA:

Dimensions and configuration: See figure 1.

Weight: 0.25 pound maximum.

Terminal strength: 3 ±0.5 pounds pull maximum.

Finish: Bright nickel per QQ-N-290, class 1, type IV.

1/ Input terminals 1, 5 and 9 must be connected together during this test.
2/ All pins to case for this test.

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(8) PART NUMBER: Consists of the prefix M83726/8-, a four-digit dash number expressed in milliseconds, and a quality level indication as follows:

Military designation	M	83726/8-	1001	W
Specification sheet number				
The first three digits are significant; the fourth digit is the number of zeros to follow the first three digits. (The time delay is expressed in milliseconds and converted to seconds.) See examples.				
Quality level indication (W, Y, or X). Refer to MIL-R-83726. 3/				

EXAMPLES: M83726/8-1000W - 100-millisecond time delay, W level
M83726/8-1001X - 1-second time delay, X level
M83726/8-3003Y - 300-second time delay, Y level

3/ Any relays numbered prior to the date of this specification without a quality level indication shall be considered interchangeable (store and issue) with the "W" quality level.

Custodians:
Navy - EC
Air Force - 85

Review activities:
Navy - EC
Air Force - 11, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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